DO NOW — RECALL

Hash tables and collisions

Name: _____ Class: _____ Date: _____

1. In an undirected graph, why does edge A–B appear in the adjacency list of both A and B?

2. In a BST, 18 is smaller than root 25. Which direction do you move?

3. What are valid indexes of a five-element C# array?

4. Calculate 17 MOD 6.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Because traversal is permitted both ways.

2. Model answer

Left.

3. Model answer

0, 1, 2, 3, 4.

4. Model answer

5, since 17 = 2×6 + 5.

Vocabulary — know these terms

hash function — Maps a key to an initial bucket index.

collision — Two keys map to the same initial bucket.

linear probing — Checks successive buckets, wrapping around when needed.

load factor — The proportion of occupied table buckets.

clustering — A sequence of occupied buckets increasing probe lengths.

NEW LEARNING

Read, annotate and explain

Calculate the index

9 MOD 7 = 2, so put 9 into bucket 2.

Detect a collision

16 MOD 7 is also 2. The bucket is occupied, so move to the next bucket, 3.

Probe until empty

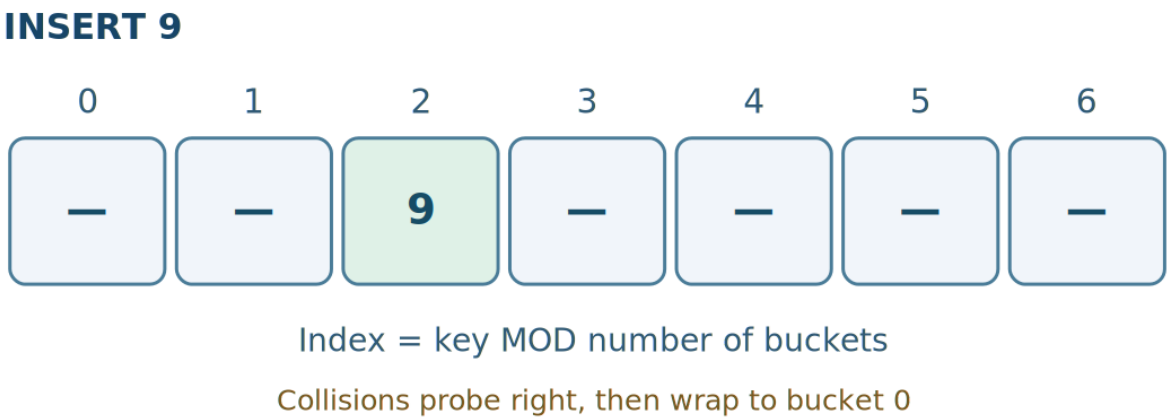
23 also hashes to 2; buckets 2 and 3 are full, so place it in 4.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

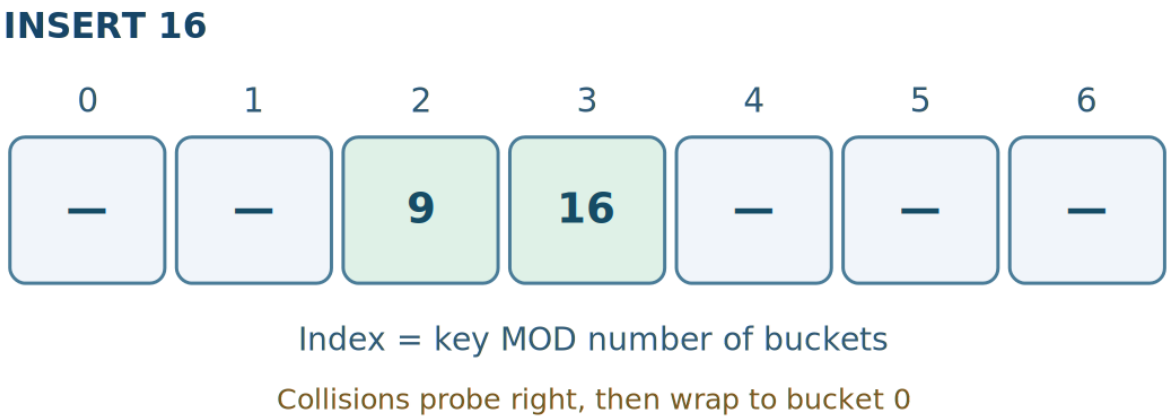
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



INSERT 9: buckets 0 to 6, values empty, empty, 9, empty, empty, empty, empty

New Learning

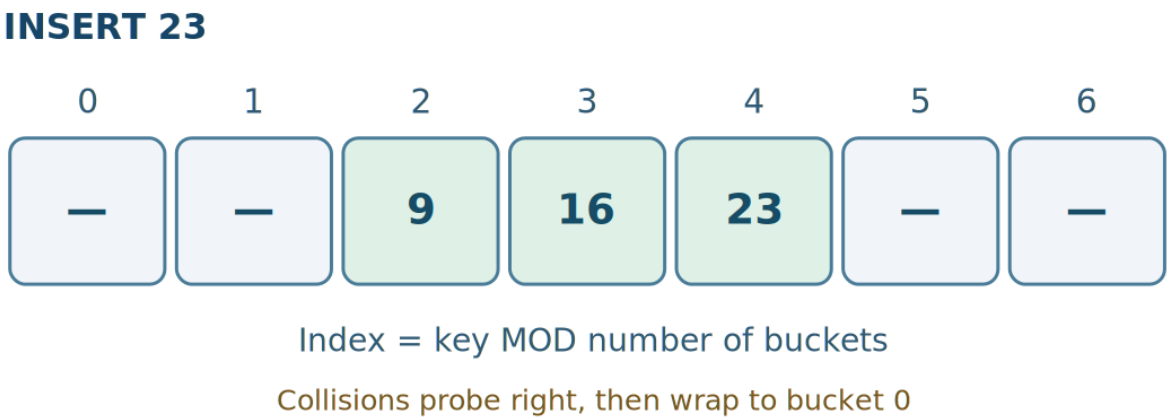


INSERT 16: buckets 0 to 6, values empty, empty, 9, 16, empty, empty, empty

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



INSERT 23: buckets 0 to 6, values empty, empty, 9, 16, 23, empty, empty

I DO — WORKED EXAMPLE

Follow the worked method

Insert 6, 11, 16 into five buckets using h(k)=k MOD 5 and linear probing.

Predict and explain

Insert 6 at index 1, because 6 MOD 5 = 1.

Check and extend

11 collides at 1, so store it at 2.

Check and extend

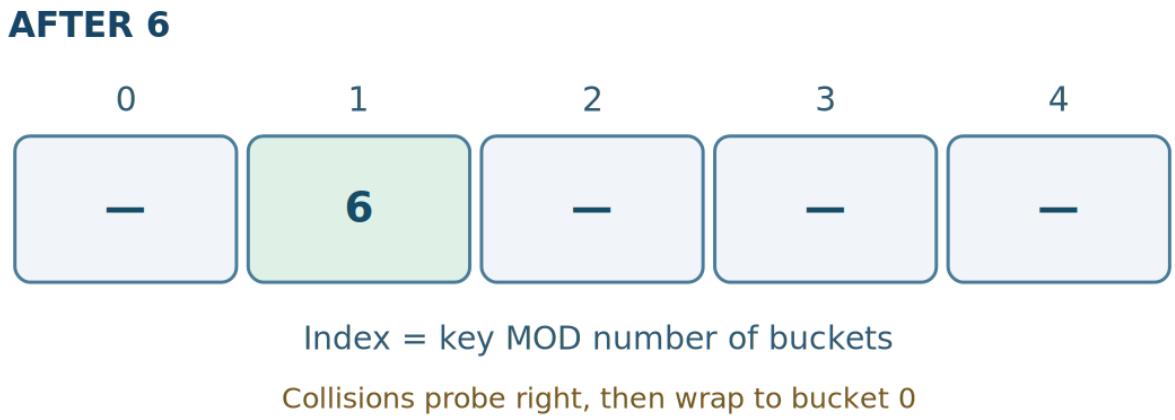
16 collides at 1 and 2, so store it at 3.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 6, 11, 16 into five buckets using h(k)=k MOD 5 and linear probing.

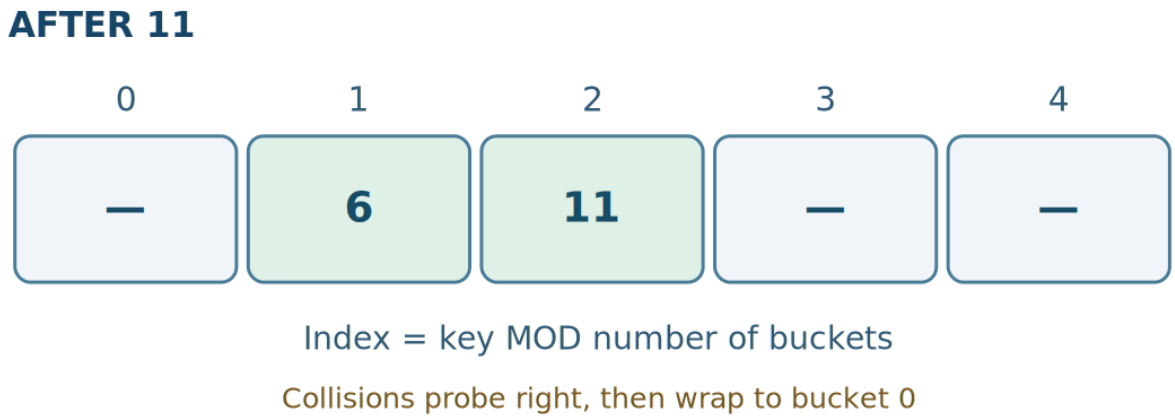
Insert 6 at index 1, because 6 MOD 5 = 1.



AFTER 6: buckets 0 to 4, values empty, 6, empty, empty, empty

Insert 6, 11, 16 into five buckets using h(k)=k MOD 5 and linear probing.

Insert 6 at index 1, because 6 MOD 5 = 1.



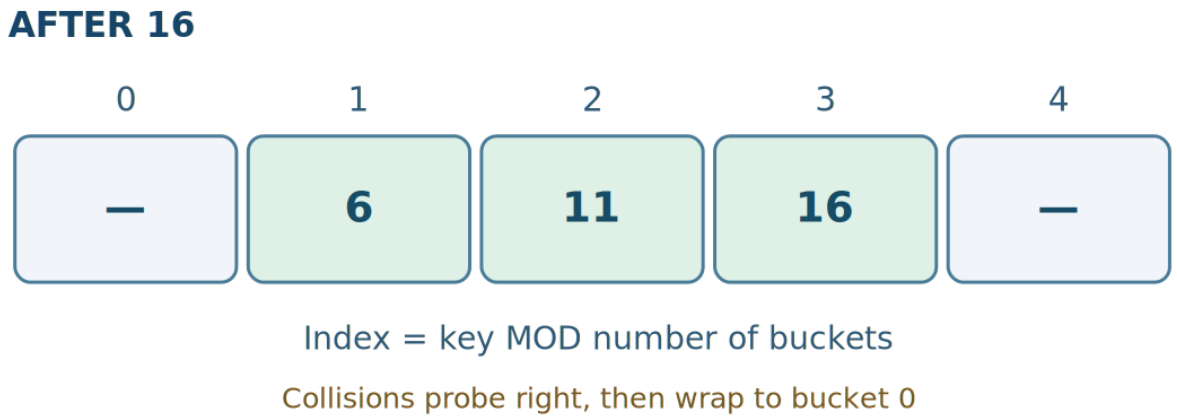
AFTER 11: buckets 0 to 4, values empty, 6, 11, empty, empty

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 6, 11, 16 into five buckets using h(k)=k MOD 5 and linear probing.

Insert 6 at index 1, because 6 MOD 5 = 1.



AFTER 16: buckets 0 to 4, values empty, 6, 11, 16, empty

WE DO — GUIDED PRACTICE

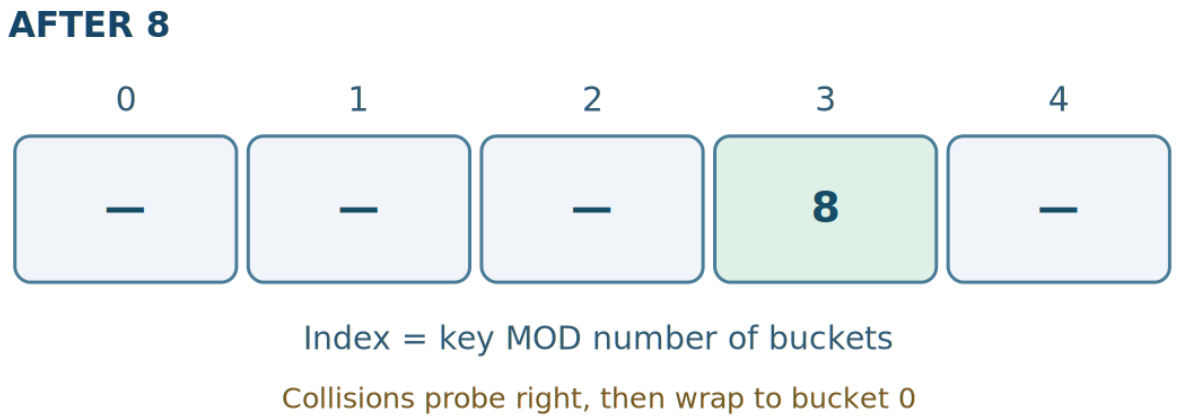
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 8, 13, 18 into five buckets using h(k)=k MOD 5.

8 goes into bucket 3.



AFTER 8: buckets 0 to 4, values empty, empty, empty, 8, empty

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict and explain 8 goes into bucket 3.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Using h(k)=k MOD 7, find the starting index for 24.

Question 2

Insert 4, 9, 14 into five empty buckets using h(k)=k MOD 5 and linear probing. Give final indexes.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain the effect of a high load factor on linear probing.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A hash collision means two different keys have the same starting bucket.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

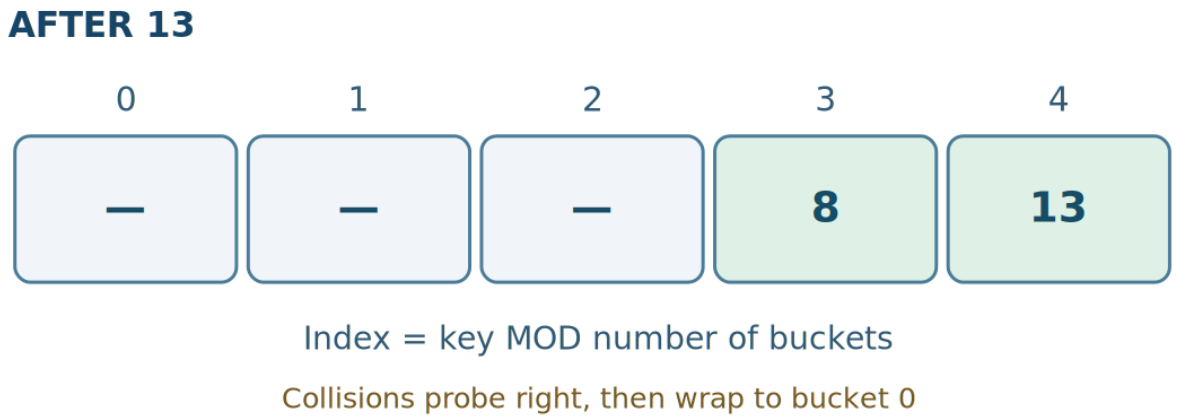
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 8, 13, 18 into five buckets using h(k)=k MOD 5.

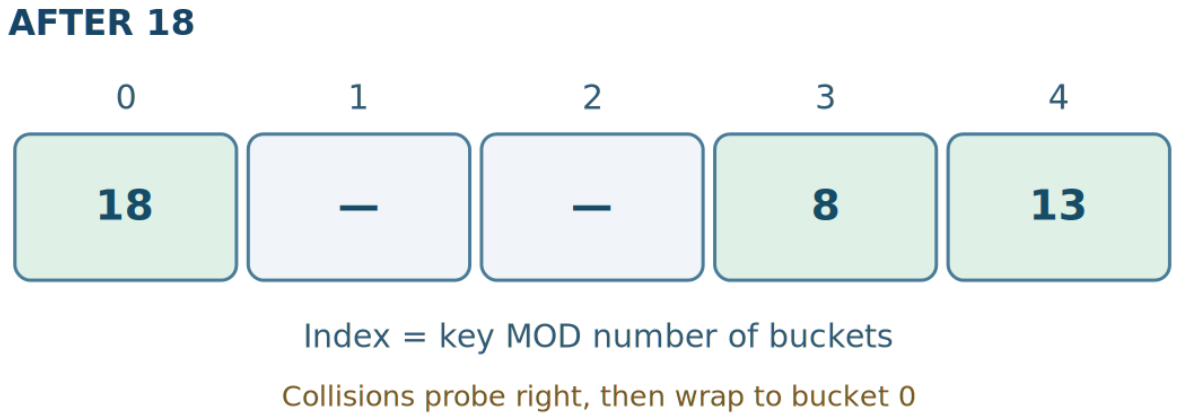
8 goes into bucket 3.



AFTER 13: buckets 0 to 4, values empty, empty, empty, 8, 13

Insert 8, 13, 18 into five buckets using h(k)=k MOD 5.

8 goes into bucket 3.



AFTER 18: buckets 0 to 4, values 18, empty, empty, 8, 13

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check and extend

13 collides at 3 and probes to 4.

18 collides at 3 and 4; linear probing wraps around to 0, which is empty, so 18 goes into bucket 0.

IMPROVE: Explain one correction to your own reasoning.